

Personal Portfolio

Complete Build Brief for Cursor AI Agent

Next.js (App Router) + Tailwind CSS + Three.js (React Three Fiber)

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How to use this PDF: Drop it into your project folder (e.g. /docs/PORTFOLIO_BRIEF . pdf) or paste its text into Cursor. Then tell the agent: *"Read PORTFOLIO_BRIEF and build Phase 1. Stop and wait for my approval before Phase 2."* Building phase by phase (Section 14) gives much better results than one giant prompt.

1. Project Overview

Build a fast, modern, single-page developer portfolio for a full-stack (MERN) developer who also runs e-commerce fashion brands. The site must feel premium and memorable (a subtle 3D hero), but still load fast and work well on mobile. Content is driven by typed data files so it can be edited without touching components.

Goals

- Attract freelance clients and employers; clear call-to-action to contact.
- Show real projects with live links, tech used, and outcome.
- Stand out with a tasteful 3D scene without hurting Lighthouse scores.
- Easy to update: all text, projects, skills live in `/data`.

Target audience

- Startup founders and business owners needing web apps / e-commerce stores.
- Recruiters and tech leads looking at full-stack ability.
- International and Pakistani clients (language: English).

Non-negotiables

- TypeScript strict mode, no any.
- Mobile-first responsive design (360px up to 1920px).
- Lighthouse targets: Performance 90+, Accessibility 95+, SEO 100, Best Practices 95+ (desktop).
- 3D must be lazy-loaded, have a fallback, and respect `prefers-reduced-motion`.
- Dark mode default with light mode toggle.

2. Tech Stack

Layer	Choice	Notes
Framework	Next.js (latest stable, App Router)	Use <code>src/ dir</code> , TypeScript, ESLint
Styling	Tailwind CSS (latest stable)	Design tokens via CSS variables / theme config
3D	three + <code>@react-three/fiber</code> + <code>@react-three/drei</code>	Client-only, dynamic import with <code>ssr:false</code>
Animation	framer-motion (UI) + optional GSAP ScrollTrigger	Keep bundle small; use one primary lib
Smooth scroll	lenis (optional)	Disable when reduced-motion is on
Theme	next-themes	class strategy, default dark
Forms	react-hook-form + zod	Validation shared client + server
Email	Resend (or Nodemailer + Gmail SMTP)	Route handler at <code>/api/contact</code>
Icons	lucide-react + react-icons (brands)	
Fonts	next/font (Inter + Space Grotesk or Sora)	No external font requests at runtime
Deploy	Vercel	Env vars in dashboard

Agent instruction: Always check the current stable versions and installation docs of Next.js, Tailwind, and React Three Fiber before installing (Tailwind setup differs between major versions). Do not guess config syntax from memory.

3. Setup Commands

Run these in order (the agent can run them in the integrated terminal):

```
npx create-next-app@latest portfolio --typescript --tailwind --eslint \
  --app --src-dir --import-alias "@/*"
cd portfolio

npm i three @react-three/fiber @react-three/drei
npm i -D @types/three

npm i framer-motion next-themes lucide-react react-icons
npm i react-hook-form zod @hookform/resolvers
npm i resend
npm i lenis

npm run dev
```

4. Folder Structure

```
src/
  app/
    layout.tsx      # fonts, ThemeProvider, metadata, JSON-LD
    page.tsx        # composes all sections
    globals.css      # tailwind + CSS variables
    api/contact/route.ts # POST: validate (zod) + send email
    sitemap.ts robots.ts opengraph-image.tsx
  components/
    layout/  Navbar.tsx Footer.tsx ThemeToggle.tsx ScrollProgress.tsx
    sections/ Hero.tsx About.tsx Skills.tsx Projects.tsx
              Experience.tsx Services.tsx Contact.tsx
    three/   HeroScene.tsx Scene.tsx FloatingShapes.tsx Fallback.tsx
    ui/      Button.tsx Section.tsx Badge.tsx Card.tsx Reveal.tsx
  data/
    profile.ts skills.ts projects.ts experience.ts services.ts
  lib/
    utils.ts (cn helper) validators.ts motion.ts
  hooks/
    useReducedMotion.ts useIsMobile.ts
  public/
    images/projects/ resume.pdf favicon + og assets
```

5. Design System

Look and feel

Clean, dark, minimal with one vivid accent. Lots of whitespace, large typography, glassy cards with subtle borders, soft glow behind the 3D object. Think modern developer-agency style, not a template.

Color tokens (define as CSS variables, map in Tailwind theme)

Token	Dark	Light
background	#0B0B12	#FAFAFC
surface (cards)	#14141F	#FFFFFF
border	rgba(255,255,255,0.08)	rgba(0,0,0,0.08)
text	#EDED7F	#14141F
muted	#9A9AB0	#5A5A6E
accent	#6D5DFB	#5B4BEB
accent-2 (gradient end)	#22D3EE	#0EA5C9

- **Typography:** Headings Space Grotesk/Sora (600-700), body Inter. Hero H1 clamp(2.5rem, 7vw, 5.5rem).
- **Container:** max-w-6xl, px-5 on mobile, px-8 on desktop. Section padding py-24 md:py-32.
- **Radius:** cards rounded-2xl, buttons rounded-full.
- **Effects:** gradient text on key words, subtle noise or grid background, hover lift on cards.
- **Buttons:** primary (accent gradient), secondary (outline), both with focus-visible ring.

6. Sections - Detailed Spec

6.1 Navbar

- Sticky, transparent at top, blur + border after scrolling 20px.
- Logo (name or monogram "U"), links: About, Skills, Projects, Experience, Contact; Theme toggle; "Hire me" button.
- Mobile: hamburger opens full-screen overlay menu with animated links.
- Active link highlight using IntersectionObserver. Smooth scroll to anchors.

6.2 Hero (with Three.js)

- Left: small label ("Full-Stack Developer - Lahore, PK"), big headline, 1-2 line intro, two CTAs (View Projects / Download CV), social icons.
- Right/background: 3D scene (Section 7). On mobile, scene sits behind text at reduced opacity or is replaced by lighter version.
- Typing/rotating text for roles: MERN Developer, Next.js Engineer, E-commerce Builder.
- Scroll-down indicator.

6.3 About

- Photo (rounded, subtle gradient border) + 2-3 short paragraphs.
- Quick-stat cards: Years of experience, Projects shipped, Brands built (numbers editable in data file; animate count-up on view).
- Mention: building and co-running e-commerce fashion brands alongside client development work, studying at Virtual University.

6.4 Skills

- Grouped grid: Frontend, Backend, Database/ORM, Tools/DevOps.
- Frontend: React, Next.js, TypeScript, Tailwind CSS, Three.js. Backend: Node.js, Express, NestJS. Database: MongoDB, MySQL, Prisma. Tools: Git, GitHub, Vercel, Postman.
- Each skill = icon + name chip with hover glow. No fake percentage bars.

6.5 Projects (most important section)

- Responsive card grid (1 / 2 columns), featured project can be larger.
- Card: cover image, title, 1-line description, tech badges, links (Live, GitHub if public).
- Click opens a modal or `/projects/[slug]` case-study page: problem, solution, features, stack, screenshots.
- Filter tabs: All / E-commerce / Web Apps / Desktop / AI.
- Hover: tilt effect (framer-motion) + image zoom.

Starting project list (edit in `data/projects.ts`):

Project	Type	Description	Stack
AI Shanaz	E-commerce	Premium women's fashion online store with SEO and growth work	Next.js, Tailwind, Node, MongoDB (adjust)
Aliya Warys	E-commerce	Premium women's fashion platform	Next.js, TypeScript (adjust)
SpareFlow	Desktop app	Offline-first Point of Sale for an auto-parts business	Desktop, offline-first DB (adjust)
AI Study Buddy	AI / Web app	Upload notes/PDFs, get summaries, flashcards, quizzes	Next.js, AI API (adjust)
SocietyPay	Web app	Maintenance fee collection for housing societies	NestJS, MySQL, Prisma (adjust)

Stack column is a starting guess - replace with the real stack for each project. Use placeholder images until real screenshots are added.

6.6 Services

- 4 cards: Custom Web Apps (MERN), E-commerce Stores, Next.js / SEO Optimization, API and Backend Development.
- Each: icon, title, 2-line description.

6.7 Experience / Education timeline

- Vertical timeline with animated line. Entries: freelance/client work, e-commerce ventures, Virtual University (ongoing), I.Com (2022).
- Content lives in `data/experience.ts`.

6.8 Contact

- Form: name, email, message (+ hidden honeypot field). Validate with zod on client and server.
- Submit to `/api/contact`, show loading, success and error states (toast).
- Side info: email, WhatsApp link, location (Lahore), GitHub / LinkedIn icons.
- Basic rate limiting on the API route (in-memory or Upstash if added later).

6.9 Footer

- Copyright, quick links, social icons, "Back to top" button.

7. Three.js / React Three Fiber Spec

Hero scene concept

A floating, slowly rotating distorted icosahedron (or torus knot) with a gradient/glass-like material, surrounded by small floating particles and 3-4 orbiting low-poly shapes. The main object subtly follows the mouse (parallax). Soft accent-colored point lights create glow.

- Use `<Canvas>` from `@react-three/fiber` with `dpr={[1, 1.5]}` and `gl={{ antialias: true, powerPreference: 'high-performance' }}`.
- Drei helpers: `Float`, `MeshDistortMaterial` or `MeshTransmissionMaterial` (only on desktop), `Environment` (preset), `Sparkles/Points`.
- Mouse parallax: lerp object rotation toward pointer in `useFrame`.
- Optional scroll-linked effect: scale/rotate scene based on scroll progress.

Performance and safety rules

- Load via `next/dynamic(() => import('./HeroScene'), { ssr: false, loading: () => <Fallback /> })`.
- Mount the Canvas only when hero is in viewport; pause rendering (`frameLoop="demand"` or `unmount`) when scrolled away.
- Mobile (width < 768px): fewer particles, no transmission material, lower dpr, or swap to CSS gradient blob.
- If `prefers-reduced-motion`: static render, no rotation or parallax.
- WebGL not supported: show CSS gradient fallback.
- No heavy external models/textures. If a .glb is used, compress with gltf-transform (Draco) and keep under 1 MB.
- Dispose geometries/materials on unmount; avoid creating objects inside `useFrame`.

```

// components/three/HeroScene.tsx (starter skeleton)
"use client";
import { Canvas, useFrame } from "@react-three/fiber";
import { Float, MeshDistortMaterial, Sparkles, Environment } from "@react-three/drei";
import { useRef } from "react";
import type { Mesh } from "three";

function Blob() {
  const ref = useRef<Mesh>(null!);
  useFrame(({ pointer }, dt) => {
    ref.current.rotation.y += (pointer.x * 0.6 - ref.current.rotation.y) * dt;
    ref.current.rotation.x += (-pointer.y * 0.6 - ref.current.rotation.x) * dt;
  });
  return (
    <Float speed={1.5} floatIntensity={1.2}>
      <mesh ref={ref}>
        <icosahedronGeometry args={[1.4, 4]} />
        <MeshDistortMaterial color="#6D5DFB" distort={0.4} speed={1.5} roughness={0.2}
          metalness={0.6} />
      </mesh>
    </Float>
  );
}

export default function HeroScene() {
  return (
    <Canvas dpr={[1, 1.5]} camera={{ position: [0, 0, 5], fov: 45 }}>
      <ambientLight intensity={0.5} />
      <pointLight position={[5, 5, 5]} intensity={20} color="#22D3EE" />
      <Blob />
      <Sparkles count={60} scale={8} size={2} speed={0.4} />
      <Environment preset="city" />
    </Canvas>
  );
}

```

8. Data Files (content layer)

All content comes from typed files in /data. Components must never hardcode personal text.

```
// data/profile.ts
export const profile = {
  name: "Usman",
  role: "Full-Stack Developer (MERN)",
  tagline: "I build fast, scalable web apps and e-commerce stores.",
  location: "Lahore, Pakistan",
  email: "YOUR_EMAIL@example.com",
  whatsapp: "https://wa.me/92XXXXXXXXXX",
  github: "https://github.com/Usmandevblends",
  linkedin: "https://linkedin.com/in/YOUR_HANDLE",
  resume: "/resume.pdf",
  roles: ["MERN Developer", "Next.js Engineer", "E-commerce Builder"],
  stats: [
    { label: "Years Experience", value: 3 },
    { label: "Projects Shipped", value: 15 },
    { label: "Brands Built", value: 2 },
  ],
} as const;

// data/projects.ts
export type Project = {
  slug: string; title: string; summary: string;
  category: "E-commerce" | "Web App" | "Desktop" | "AI";
  image: string; tech: string[]; live?: string; repo?: string;
  featured?: boolean; problem?: string; solution?: string; features?: string[];
};
export const projects: Project[] = [ /* fill with the list from Section 6.5 */ ];
```

9. Animation Guidelines

- Reusable <Reveal> component: fade + 24px slide-up on scroll into view, once: true.
- Stagger children in grids (0.08s delay).
- Hero text: line-by-line reveal on load.
- Durations 0.4-0.7s, easing easeOut. Nothing flashy or looping except the 3D scene.
- Wrap everything with a reduced-motion check.

10. SEO, Accessibility, Performance

- Metadata API in layout.tsx: title, description, keywords, canonical, Open Graph, Twitter card; dynamic OG image.
- JSON-LD Person schema. sitemap.ts and robots.ts.
- Semantic HTML (header, main, section with aria-labelledby, footer). One H1.
- All images via next/image with width/height + alt text; AVIF/WebP.
- Keyboard navigable, visible focus rings, color contrast AA, skip-to-content link.
- Lazy-load below-the-fold sections and the 3D chunk; analyze bundle with @next/bundle-analyzer.
- Add Vercel Analytics (optional).

11. Contact API Spec

```
// app/api/contact/route.ts
import { NextResponse } from "next/server";
import { z } from "zod";
import { Resend } from "resend";

const schema = z.object({
  name: z.string().min(2).max(80),
  email: z.string().email(),
  message: z.string().min(10).max(2000),
  website: z.string().max(80).optional(), // honeypot - must be empty
});

export async function POST(req: Request) {
  const parsed = schema.safeParse(await req.json());
  if (!parsed.success) return NextResponse.json({ error: "Invalid data" }, { status: 400 });
  if (parsed.data.website) return NextResponse.json({ ok: true }); // bot

  const resend = new Resend(process.env.RESEND_API_KEY);
  await resend.emails.send({
    from: "Portfolio <onboarding@resend.dev>",
    to: process.env.CONTACT_TO_EMAIL!,
    replyTo: parsed.data.email,
    subject: `New message from ${parsed.data.name}`,
    text: parsed.data.message,
  });
  return NextResponse.json({ ok: true });
}
```

Environment variables (.env.local, never commit):

```
RESEND_API_KEY=
CONTACT_TO_EMAIL=
NEXT_PUBLIC_SITE_URL=https://yourdomain.com
```

12. Deployment

- Push to GitHub, import repo into Vercel, add env vars, deploy.
- Add custom domain, enable HTTPS (automatic).
- Run Lighthouse on the deployed URL and fix anything under target.
- Submit sitemap to Google Search Console.

13. Cursor Rules (save as .cursor/rules or .cursorrules)

You are building a Next.js (App Router) + TypeScript + Tailwind + React Three Fiber portfolio.

- Follow PORTFOLIO_BRIEF.pdf strictly. Ask before deviating.
- TypeScript strict. No `any`. Prefer server components; add "use client" only when needed.
- All content comes from /src/data. No hardcoded personal text in components.
- Three.js code lives only in /src/components/three and is dynamically imported (ssr:false).
- Respect prefers-reduced-motion and mobile performance rules.
- Use Tailwind utilities + CSS variables for theme tokens. No inline styles unless dynamic.
- Check latest docs for Next.js / Tailwind / R3F before configuring. Do not guess APIs.
- After each phase: run `npm run lint` and `npm run build`, fix errors, then summarize what was done.
- Keep components small, accessible, and typed. Reuse ui/ primitives.

14. Build Phases (give these to Cursor one at a time)

Phase	Task	Done when
1	Project setup, folders, fonts, theme tokens, ThemeProvider, ui/ primitives (Button, Section, Badge, Card, Reveal), data files with placeholders	App runs, dark/light toggle works, build passes
2	Navbar (desktop + mobile menu), Footer, scroll progress, smooth scroll	Navigation + active section highlight work
3	Hero section text + animations (no 3D yet)	Hero looks finished and responsive
4	Three.js hero scene, lazy loading, fallback, mobile + reduced-motion handling	60fps on desktop, graceful on mobile
5	About + Skills sections	Content from data files, count-up stats work
6	Projects grid, filters, hover effects, case-study pages /projects/[slug]	All projects render, links work
7	Services + Experience timeline	Responsive and animated
8	Contact form + /api/contact + validation + toasts	Email arrives; errors handled
9	SEO: metadata, OG image, JSON-LD, sitemap, robots; accessibility pass	Lighthouse targets met
10	Polish, bundle analysis, cross-browser/mobile testing, deploy to Vercel	Live URL, no console errors

Ready-to-paste prompt for Phase 1

```
Read PORTFOLIO_BRIEF.pdf fully. Execute Phase 1 only:
set up the Next.js + Tailwind project per Section 3 and 4, create the design tokens
from Section 5, ThemeProvider (dark default), the ui/ primitives, and the /data files
with placeholder content. Run lint and build. Then stop and summarize what you did
and what Phase 2 will do. Do not start Phase 2.
```

15. Final Acceptance Checklist

- ☐ All 10 phases complete, npm run build passes with zero errors/warnings.
- ☐ Looks right at 360px, 768px, 1280px, 1920px.
- ☐ 3D scene lazy-loaded, has fallback, disabled/reduced on mobile + reduced-motion.
- ☐ Contact form sends a real email and handles errors.
- ☐ Lighthouse: Perf 90+, A11y 95+, SEO 100, Best Practices 95+.
- ☐ No hardcoded personal text in components; all in /data.
- ☐ Real resume.pdf, real project screenshots, real social links added.
- ☐ Deployed on Vercel with custom domain.

16. Things YOU Must Fill In

- Email, WhatsApp, LinkedIn, profile photo, resume PDF, domain, Resend API key.
- Real project screenshots, live URLs, accurate tech stacks, and real stats.